PVsyst V8.0.21

PVsyst - Simulation report

Grid-Connected System

Project: Solar Lounge

Variant: New simulation variant

Tables on a building

System power: 13.60 kWp

[redacted] - Israel

PVsyst TRIAL

Author

PVsyst V8.0.21
VC0, Simulation date:
06/04/26 19:28
with V8.0.21

Project summary

Geographical Site
[redacted]
Israel

Situation
Latitude [redacted]
Longitude [redacted]
Altitude 292 m
Time zone UTC+2

Project settings
Albedo 0.20

Weather data
[redacted]
PVGIS api TMY

System summary

Grid-Connected System

Tables on a building

Orientation #1
Fixed plane
Tilt/Azimuth 26 / 0.2 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Daily household consumers
Constant over the year
Average 45 kWh/Day

System information

PV Array
Nb. of modules 20 units
Pnom total 13.60 kWp

Inverters
Nb. of units 1 unit
Total power 10 kWac
Pnom ratio 1.36

Battery pack
Storage strategy: Self-consumption
Nb. of units 3 units
Voltage 154 V
Capacity 93 Ah

Results summary

Produced Energy	23443 kWh/year	Specific production	1724 kWh/kWp/year	Perf. Ratio PR	78.87 %
Used Energy	16542 kWh/year			Solar Fraction SF	56.63 %

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Near shading definition - Iso-shadings diagram 5
Detailed User's needs 6
Main results 7
Loss diagram 8
Predef. graphs 9
Single-line diagram 10
CO_2 Emission Balance 11

PVsyst TRIAL

PVsyst V8.0.21
VC0, Simulation date:
06/04/26 19:28
with V8.0.21

General parameters

Grid-Connected System

Tables on a building

Orientation #1
Fixed plane
Tilt/Azimuth 26 / 0.2 °

Models used
Transposition Perez
Diffuse Imported
Circumsolar separate

Horizon
Free Horizon

Near Shadings
Linear shadings : Fast (table)

Storage
Kind Self-consumption

User's needs
Daily household consumers
Constant over the year
Average 45 kWh/Day

PV Array Characteristics

PV module
Manufacturer Generic
Model TSM-NEG-21C-20-680 Vertex
(Custom parameters definition)
Trina_Vertex_N_680.PAN
Unit Nom. Power 680 Wp
Number of PV modules 20 units
Nominal (STC) 13.60 kWp
Modules 2 string x 10 In series
At operating cond. (50°C)
Pmpp 12.61 kWp
U mpp 365 V
I mpp 35 A

Inverter
Manufacturer Generic
Model S6-EH3P10K2-H
(Original PVsyst database)
Unit Nom. Power 10.00 kWac
Number of inverters 1 unit
Total power 10.0 kWac
Operating voltage 200-850 V
Pnom ratio (DC:AC) 1.36
Power sharing within this inverter

Total PV power
Nominal (STC) 14 kWp
Total 20 modules
Module area 62.1 m²
Cell area 58.2 m²

Total inverter power
Total power 10 kWac
Number of inverters 1 unit
Pnom ratio 1.36

Grid storage - Self-consumption

Battery
Manufacturer Generic
Model Adjustable Li-Ion

Battery pack
Nb. of units 3 in series
Voltage 154 V
Nominal Capacity (C10) 93 Ah
Nominal energy 14.3 kWh
Operating SOC limits 95.0 / 20.0 %
Useable energy 10.8 kWh
Operating temperature Fixed 20 °C

Battery input charger
Model Generic
Max. charging power 7.00 kWdc
Max./Euro effic. 97.0 / 95.0 %

Battery to Grid inverter
Model Generic
Max. discharging power 5.00 kWac
Max./Euro effic. 97.0 / 95.0 %

PVsyst TRIAL

PVsyst V8.0.21
VC0, Simulation date:
06/04/26 19:28
with V8.0.21

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 171 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.38 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

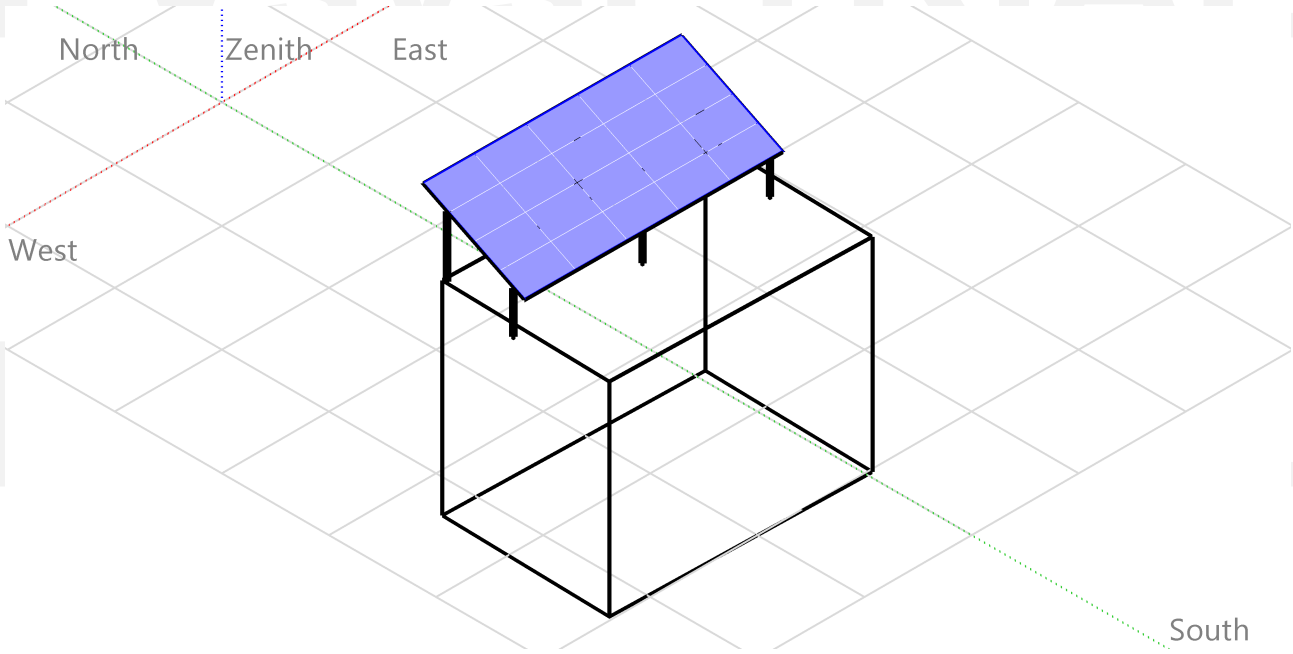
IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

PVsyst TRIAL

Near shadings parameter

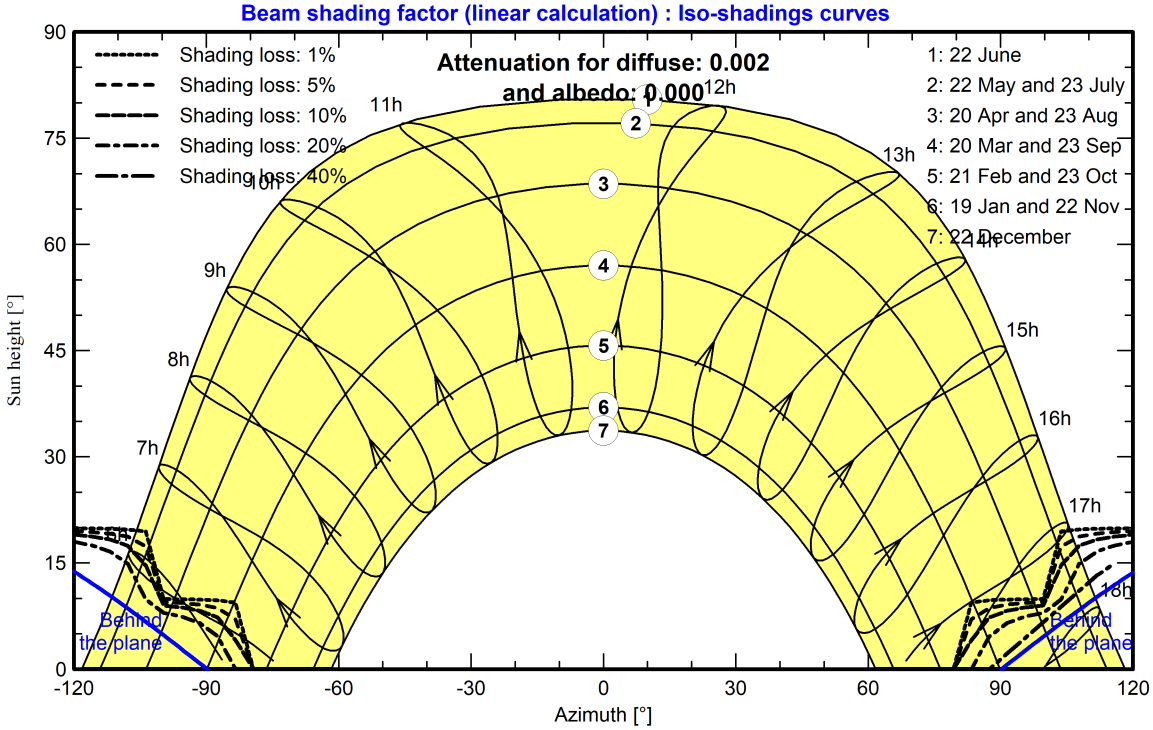
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 26°/ 0.2°

Beam shading factor (linear calculation) : Iso-shadings curves



PVsyst V8.0.21
VC0, Simulation date:
06/04/26 19:28
with V8.0.21

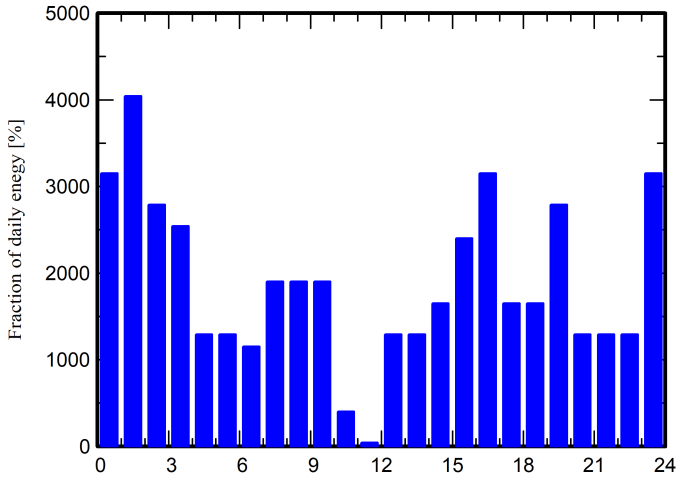
Detailed User's needs

Daily household consumers, Constant over the year, average = 45 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	30	12/lamp	12.0	4320
TV / PC / Mobile	10	150/app	10.0	15000
Fridge / Deep-freeze	2		24	1000
Other uses	1	2500 tot	10.0	25000
Total daily energy				45320

Hourly distribution



PVsyst TRIAL

PVsyst V8.0.21
VC0, Simulation date:
06/04/26 19:28
with V8.0.21

Main results

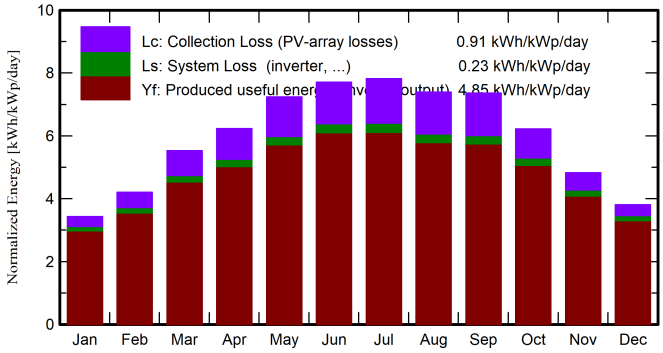
System Production

Produced Energy	23443 kWh/year	Specific production	1724 kWh/kWp/year
Used Energy	16542 kWh/year	Perf. Ratio PR	78.87 %
		Solar Fraction SF	56.63 %

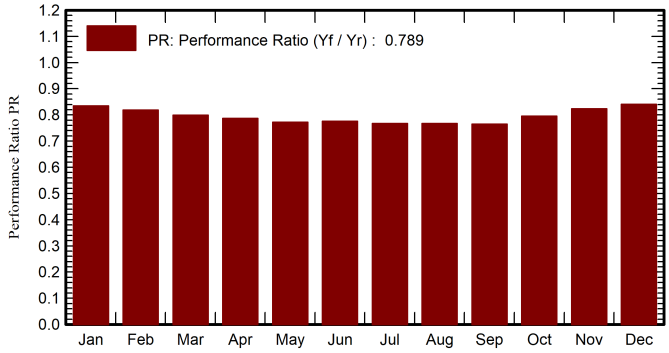
Battery ageing (State of Wear)

Cycles SOW	93.3 %
Static SOW	90.0 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_User kWh	E_Solar kWh	E_Grid kWh	EFrGrid kWh
January	74.9	34.63	11.10	106.3	104.1	1318	1405	660	546	745
February	93.0	41.76	10.89	117.7	115.4	1419	1269	613	697	656
March	147.1	59.15	14.06	171.3	168.0	2002	1405	772	1089	633
April	178.0	69.25	16.34	186.8	182.9	2148	1360	811	1188	548
May	229.4	70.75	22.24	224.4	219.8	2526	1405	879	1477	526
June	247.8	62.47	23.75	231.4	226.1	2611	1360	884	1557	475
July	254.5	60.77	26.83	242.3	237.0	2704	1405	912	1617	493
August	223.0	62.79	27.59	229.1	224.8	2558	1405	884	1506	521
September	193.5	50.14	25.41	220.8	216.5	2456	1360	828	1467	532
October	150.3	48.32	22.04	192.8	189.4	2238	1405	789	1296	616
November	102.2	39.28	17.39	144.7	142.1	1750	1360	691	930	669
December	79.2	34.17	11.49	118.0	115.6	1464	1405	644	704	761
Year	1973.1	633.49	19.14	2185.4	2141.7	25192	16542	9367	14076	7175

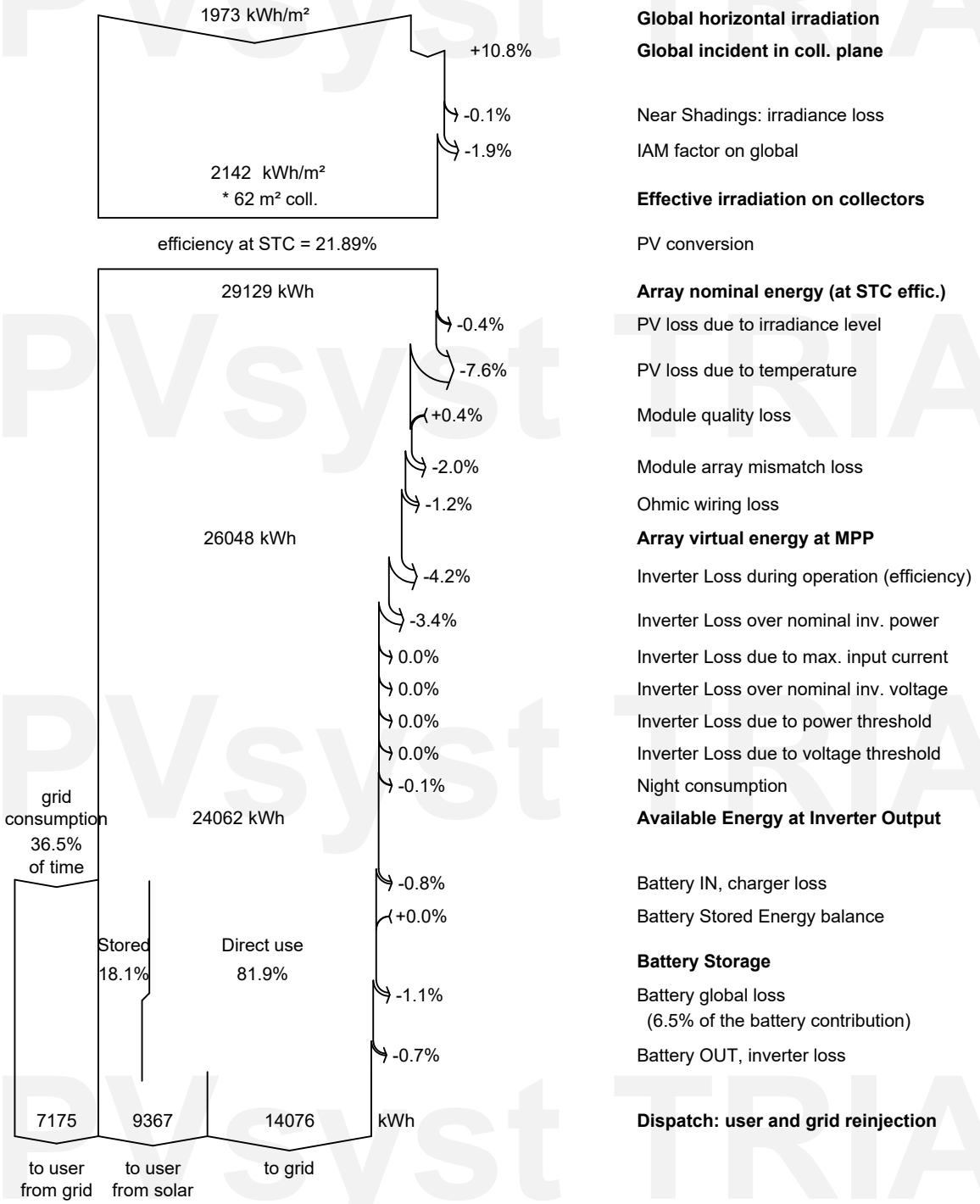
Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_User	Energy supplied to the user
T_Amb	Ambient Temperature	E_Solar	Energy from the sun
GlobInc	Global incident in coll. plane	E_Grid	Energy injected into grid
GlobEff	Effective Global, corr. for IAM and shadings	EFrGrid	Energy from the grid

PVsyst TRIAL

PVsyst V8.0.21
VC0, Simulation date:
06/04/26 19:28
with V8.0.21

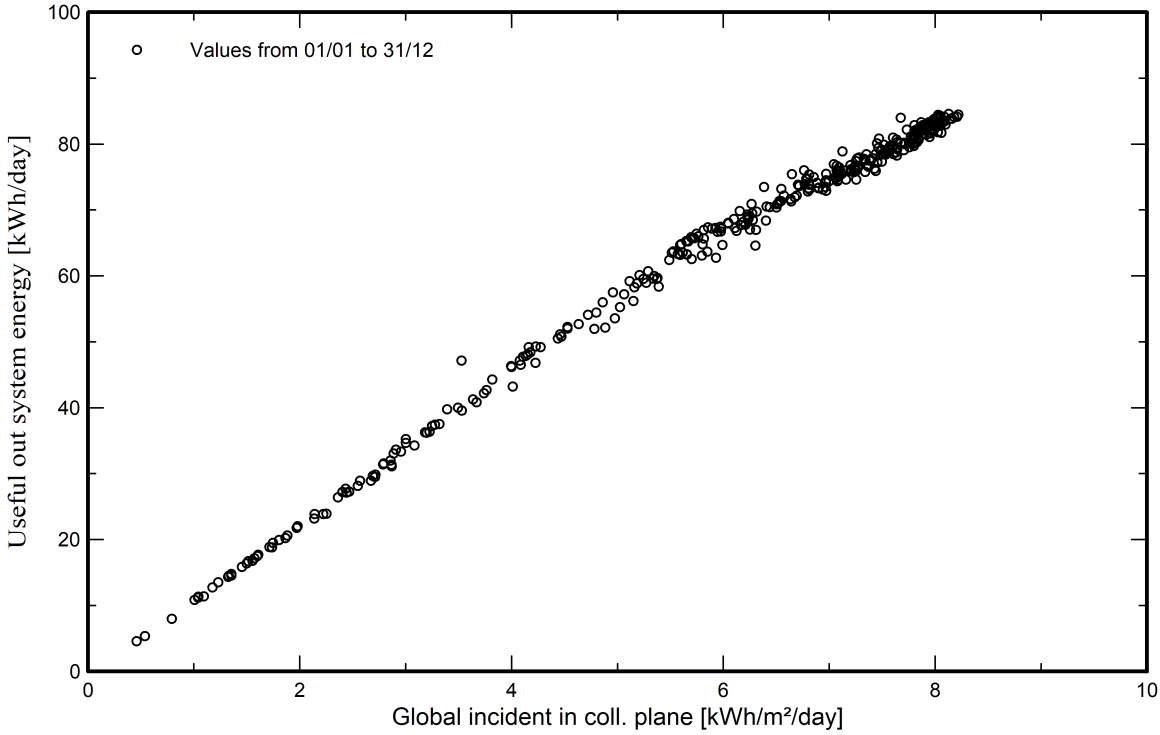
Loss diagram



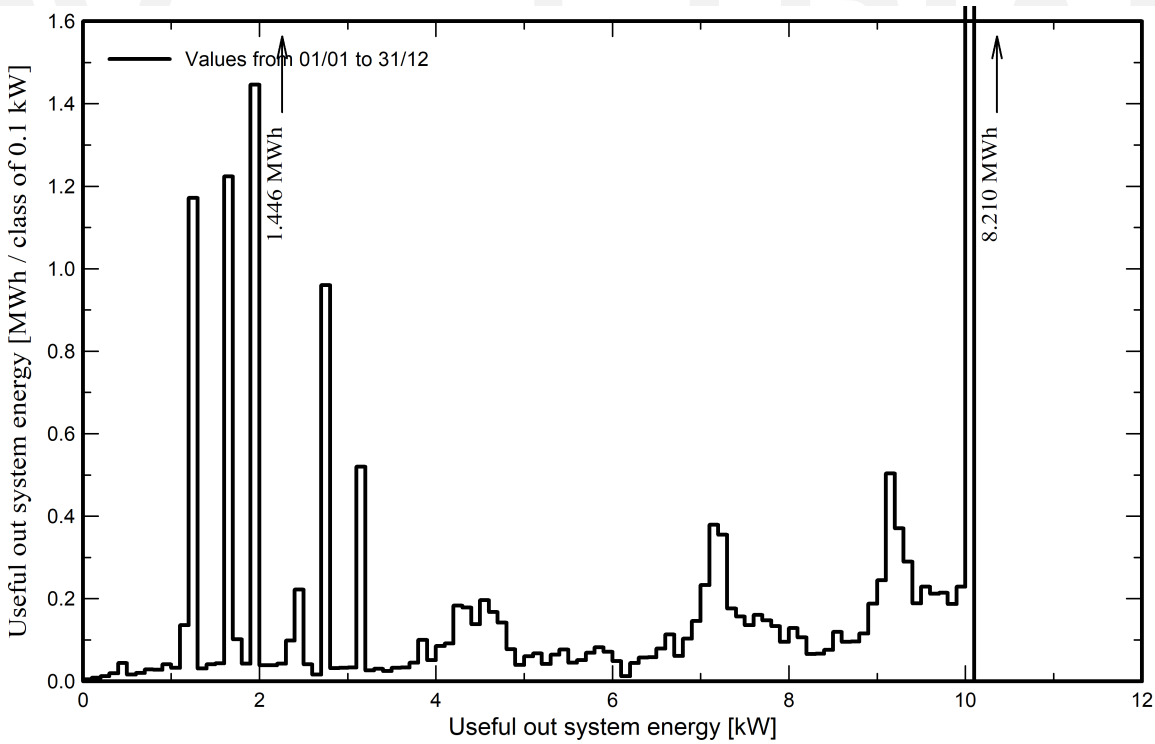
PVsyst V8.0.21
VC0, Simulation date:
06/04/26 19:28
with V8.0.21

Predef. graphs

Daily Input/Output diagram

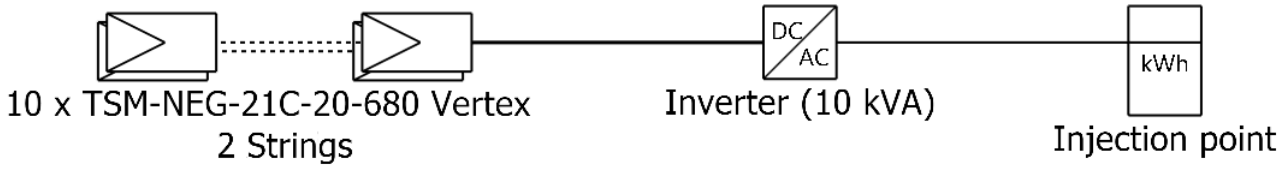


System Output Power Distribution



Single-line diagram

PVsyst V8.0.21
VC0, Simulation date:
06/04/26 19:28
with V8.0.21



PV module	TSM-NEG-21C-20-680 Vertex
Inverter	S6-EH3P10K2-H
String	10 x TSM-NEG-21C-20-680 Vertex

Solar Lounge	
VC0 : New simulation variant	06/04/26

PVsyst V8.0.21
VC0, Simulation date:
06/04/26 19:28
with V8.0.21

CO_2 Emission Balance

Total:	419.2 tCO_2

Generated emissions

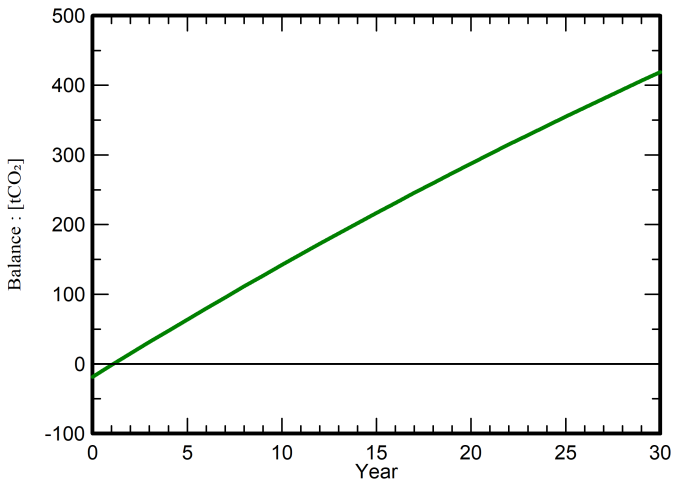
Total:	18.63 tCO_2

Source: Detailed calculation from table below

Replaced Emissions

Total:	504.6 tCO_2
System production:	24.06 MWh/yr
Grid Lifecycle Emissions:	699 gCO_2/kWh
Source:	IEA List
Country:	Israel
Lifetime:	30 years
Annual degradation:	1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1713 kgCO2/kWp	10.2 kWp	17470
Supports	4.66 kgCO2/kg	150 kg	699
Inverters	462 kgCO2/	1.00	462

PVsyst TRIAL